ADR Lab – Cybersecurity Project Catalog

This catalog contains 15 industry-ready Cybersecurity projects suitable for internships, final year projects, and production-level implementations.

1. AI-Powered Phishing Email Detection System

Features

- Classifies emails as phishing/legitimate
- Analyzes email text, sender address, URLs
- Detects suspicious patterns using ML
- Dashboard with detection statistics

Tech Stack

Python, Scikit-Learn/TensorFlow, NLP (spaCy), Flask, HTML/CSS

2. Intrusion Detection System (IDS) using Machine Learning

Features

- Monitors network logs
- Detects anomalies like DoS, brute force, port scan
- Real-time alert generation
- Visualization dashboards

Tech Stack

Python, Pandas, Scikit-Learn, CIC-IDS/KDD99 dataset, Matplotlib

3. Web Application Vulnerability Scanner

Features

- Crawl webpages
- Detect SQL Injection, XSS, CSRF, broken auth
- Find insecure headers & server misconfigurations
- Generates security report

Tech Stack

Python, BeautifulSoup, Requests, OWASP Top-10 rules

4. Malicious URL Detection Using Machine Learning

Features

- Analyze URL Structure and domain features
- Classifies URL as safe or malicious
- Detects phishing pages
- Simple UI to input URLs

Tech Stack

Python, Scikit-Learn, URL datasets, Flask/Streamlit

5. Secure File Storage System with Encryption

Features

- Encrypt files using AES-256
- User authentication & role-based access
- Secure download/upload
- Logs all user activities

Tech Stack

Python/Node.js, AES library (PyCryptodome), MySQL/MongoDB

6. Honeypot & Attack Analysis Dashboard

Features

- Deploy SSH/HTTP honeypot

- Logs attacker IPs & commands
- Generates attack frequency charts
- Geolocation of attacks

Tech Stack

Cowrie Honeypot, Python, ELK Stack (Elasticsearch-Logstash-Kibana)

7. Browser Extension for Detecting Unsafe Websites

Features

- Warns users about phishing/malicious sites
- Checks SSL certificate & URL reputation
- Blocks insecure forms
- Popup alerts

Tech Stack

JavaScript, Chrome Extension API, HTML/CSS

8. Network Packet Sniffer & Analyzer

Features

- Captures packets in real-time
- Displays TCP/UDP/HTTPS/DNS data
- Detect ARP spoofing, MITM attempts
- Charts for traffic analysis

Tech Stack

Python, Scapy, Wireshark dataset, Flask

9. Ransomware Behaviour Simulator (Safe)

Features

- Encrypts test files only to simulate attack
- Shows ransomware-like warning screen
- Decrypts with key
- Logs file access & encryption patterns

Tech Stack

Python, AES Encryption, Tkinter GUI

10. Biometric Authentication System

Features

- Login using face/voice/fingerprint
- Liveness detection
- Multi-factor authentication
- Secure storage of biometric data

Tech Stack

Python, OpenCV, TensorFlow/Keras, SQLite/MySQL

11. Secure Chat App with End-to-End Encryption

Features

- Encrypted messaging using AES/RSA
- Secure key exchange
- No message stored in server
- Optional group chats

Tech Stack

Python/Node.js, Socket.IO, RSA/AES cryptography

12. Blockchain-Based Voting System

Features

- Immutable transaction log
- Secure voter registration
- Double-voting prevention
- Transparent result verification

Tech Stack

Solidity (Ethereum), Web3.js, Ganache, React/Node.js

13. Password Manager with Encrypted Vault

Features

- Stores passwords in encrypted vault
- Master password-based access
- Auto password generator
- Clipboard auto-clear

Tech Stack

Python/Java, AES-256, SQLite/MySQL, Tkinter/JavaFx

14. Log Monitoring & Alerting System (Mini SIEM)

Features:

- Collects system logs (windows/Linux)
- Correlation rules to detect attacks
- Live alerts (email/SMS)
- Dashboard analysis

Tech Stack
Python, Syslog, Elasticsearch/kibana, Bash for log collection

15. IoT Security Scanner

Features:

1. Scans IoT devices for:

   • Weak/default passwords
   • open ports
   • Unencrypted traffic
   • Firmware vulnerabilities
   • Docker Deployment

2. Generates device security score

Tech Stack

Python, Nmap API, MQTT/HTTP sniffing tools, Flask